

Python Operators: A Reference

Comparison and logical operators are used to evaluate conditions in Python. They form the foundation of decision-making loops and **if** statements, always evaluating to either **True** or **False** (Booleans).

1. Comparison Operators

Used to compare two values. Python evaluates the expression left-to-right and returns a Boolean result.

Operator	Instruction & Core Mechanics	Step-by-Step Code & Output
== (Equal to)	Checks if the values on both sides are exactly equal. Note the double equals sign, distinct from variable assignment (=).	<pre>>>> x = 10 >>> y = 10 >>> x == y True >>> x == 5 False</pre>
!= (Not equal to)	Evaluates to True if the value on the left is different from the value on the right.	<pre>>>> choice = "yes" >>> choice != "no" True >>> 5 != 5 False</pre>
> (Greater than)	Returns True if the left value is strictly larger than the right value.	<pre>>>> score = 85 >>> score > 50 True >>> 50 > 50 False</pre>
< (Less than)	Returns True if the left value is strictly smaller than the right value.	<pre>>>> balance = -5 >>> balance < 0 True >>> 10 < 5 False</pre>
>= (Greater than or equal)	Returns True if the left value is either larger than OR exactly equal to the right value.	<pre>>>> age = 18 >>> age >= 18 True >>> 20 >= 18 True</pre>
<= (Less than or equal)	Returns True if the left value is either smaller than OR exactly equal to the right value.	<pre>>>> speed = 70 >>> speed <= 70 True >>> speed <= 60 False</pre>

2. Logical Operators

Used to combine or invert multiple conditional expressions.

Operator	Instruction & Core Mechanics	Step-by-Step Code & Output
and	Returns True only if both the left and right conditions evaluate to True . If either is false, the whole thing is false.	<pre>>>> age = 25 >>> has_id = True >>> age >= 18 and has_id == True True >>> age = 16 # Change age >>> age >= 18 and has_id == True False</pre>
or	Returns True if at least one of the conditions is True . It only returns False if both statements are false.	<pre>>>> is_weekend = True >>> is_holiday = False >>> is_weekend or is_holiday True >>> # Only False if both are False >>> False or False False</pre>
not	Reverses or inverts the Boolean value that follows it. Turns True into False , and vice versa.	<pre>>>> logged_in = False >>> not logged_in # "Is not logged in?" True >>> has_access = True >>> not has_access False</pre>